

GOSIM

GOSIM Paris 2026

GOSIM Paris 2026

May 5-6, 2026 Station F, Paris



Linear Attention

Past, Present and Future

Global Open-Source Innovation Meetup

Feihu Tang@MoonshotAI

Yet Another Dev. Rel.

May. 6th, 2026

01

The Predicament of Attention Mechanisms and The Birth of Linear Attention

Explore the limitations of Transformer attention and how linear attention breaks computational bottlenecks through mathematical innovation

02

Core Principles of Linear Attention

Reveal the mathematical principles and algorithmic mechanisms behind linear attention

03

The Evolution of Linear Attention

Track important models and milestones from the birth to maturity of linear attention

04

Performance Comparison & Application Practice

Let data and case studies speak, demonstrating the real-world performance of linear attention

05

Future Outlook & Challenges

Look ahead to the future development directions and challenges facing linear attention

MinakoKojima

- ACM/ICPC Gold Medal
- Google DIA Infra Team
- Trae Expert
- Moonshot AI
 - Post-training
 - Infra
 - Dev Rel



CHAPTER 01

The Predicament of The Birth of Linear Attention

Explore the limitations of Transformer attention and how linear attention breaks computational bottlenecks through mathematical innovation

Complexity Reduction

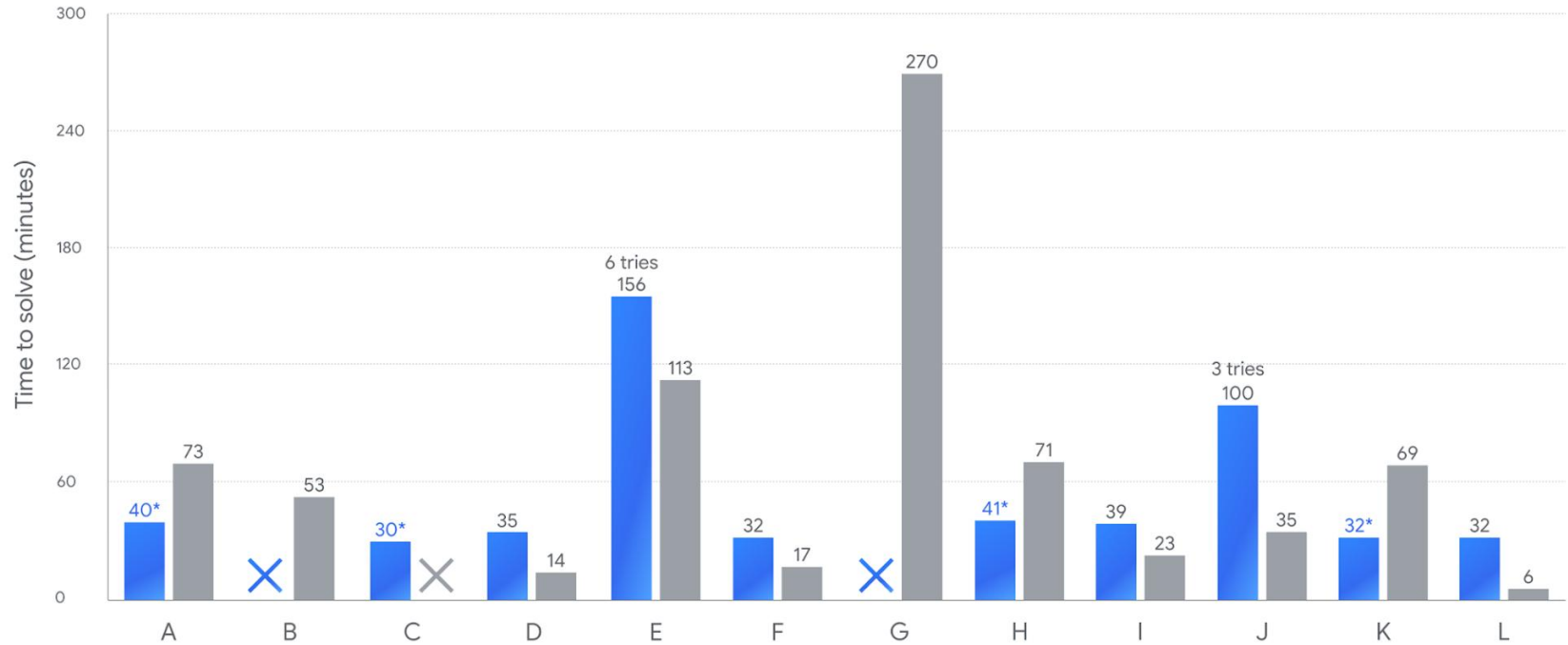
$O(N)$

Rank	Name	Solved	Time
1	 St. Petersburg State University	11	1478
	2-255 2-53 C 1-37 3-150 1-20 3-298 2-195 1-42 2-81 3-128 1-19		
2	 The University of Tokyo	10	1116
	1-208 1-271 C 1-25 1-184 1-34 G 1-95 1-66 2-72 1-126 1-17		
3	 Beijing Jiaotong University	10	1425
	2-225 2-204 1-290 1-57 2-270 2-52 G 2-132 2-84 2-107 1-148 1-6		
4	 Tsinghua University	9	865
	3-139 B 1-225 1-33 1-174 2-27 1-299 1-71 1-87 2-84 1-157 1-13		
5	 Peking University	9	887
	1-180 B C 1-14 2-168 1-50 G 1-113 1-57 1-76 1-183 1-26		
6	 Harvard University	9	995
	1-124 B C 1-35 1-238 4-78 G 2-190 1-62 1-101 1-69 1-18		
7	 University of Zagreb	9	1075
	1-15 1-242 2-77 G 3-175 2-64 1-64 1-118 2-24		
	 Massachusetts Institute of Technology	9	1123
	1-291 C 1-14 1-113 1-36 G 6-204 4-121 1-145 2-27		
	 University of Science and Technology of China	9	1128
	C 1-41 1-286 1-31 2-24 1-14		
	 National University of Science and Technology	9	1115
	3-282 C 3-259 2-238		
	 University of Novi Sad	9	1115
	1-297 C 2-259 2-238		

● Gemini

● Fastest university team

*Faster than university team



Unless marked, Gemini passed the problem with 1 try.



TEAM	A	B	C	D	E	F	G	H	I	J	K	L
OpenAI	88 1 try	104 1 try	56 1 try	23 1 try	58 1 try	40 1 try	241 9 tries	60 1 try	58 1 try	74 1 try	70 1 try	44 1 try





tfg

36 hours ago, [hide](#) <#> |

▲ +59 ▼

Can you share a prompt that solves G? I couldn't do it with GPT.

→ [Reply](#)



36 hours ago, [hide](#) <#> [^](#) |

▲ +28 ▼

[here with GPT 5 Pro](#)

I retyped the definition which includes exponentiation, but the remaining parts are copy-pasted.

→ [Reply](#)



以下の問題を解くプログラムを C++ にて作成してください。

For a positive integer a , we define a recurrence $\{b_r\}$

We say that a positive integer a

is m

-tettrative if the sequence b

stabilizes to 1

modulo m

, that is, there exists $N \geq 0$

such that $b_n \equiv 1 \pmod{m}$

for all $n \geq N$

.

For a given m

, calculate the density of the m

-tettrative integers. Here, the density of a set S

is the limit

$\lim_{n \rightarrow \infty} |S \cap [1, 2, \dots, n]| / n$.

Informally, it is the "proportion" of positive integers

-tettrative.



给 ChatGPT 发送消息

📎 附加

🔍 搜索

📖 学习

🗣️ 语音

を計算（逆元はフェルマで）。

すべての分解は 10^6 までの最小素因数表（SPF）で高速に行えます。

計算量はテストケースあたり実質「 x, y, z と $p - 1$ 達の分解」だけなのでとても軽いです。

3. 実装 (C++17, long long 使用)

cpp

复制代码

```
#include <bits/stdc++.h>
using namespace std;

using int64 = long long;
const int MAXA = 1000000;
const int64 MOD = 998244353;

// ----- SPF（最小素因数）を線形ふるいで作る -----
vector<int> spf(MAXA + 1);
vector<int> primes_list;
```

给 ChatGPT 发送消息

附加

搜索

学习

语音



Qingyu
@qingyu_shi_



显示翻译

CCPC Finals 2025 was concluded yesterday, and it's my 4th time to create the problem-set and serve as the judge for the CCPC Finals.

We the judges are quite surprised by the performance of the LLMs: ChatGPT 5.5 Pro solved ALL the tasks without any hints/assurances, which easily surpassed the strongest human teams during the onsite event.

谁会赢?

Ready when you are.

Write a C++ code to solve the following problem.

+ x Pro v

Pro thinking effort

Standard ✓

Extended



取得冠军的人类队伍

提示词都懒得写的大语言模型

第十一届中国大学生程序设计竞赛总决赛



正式赛

2026年4月26日

本试题集包含 13 道题目，共 18 页。

编号	英文标题	中文标题
A	White Night	白夜
B	Yearning for Yonder 2	对远方的向往 2
C	Four Kuhn's Theorem 2	四次伯氏定理 2
D	Call You With Your Name	呼唤你
E	Call You With Your Name 2	呼唤你 2
F	Many Many Heads 2	很多很多头 2
G	Puzzle: The Artisan of Glimoth	谜题：格里莫斯工匠
H	Strange Sorting 2	奇怪的排序 2
I	Not a Work of 1462 2	不是埃德蒙的大作 2
J	Slay the Spider 1	击败蜘蛛 1
K	Swampen Operation	沼泽操作
L	Spotted Army	斑隐
M	Rectangle Flip 1	矩形翻转 1

China Collegiate Programming Contest Finals



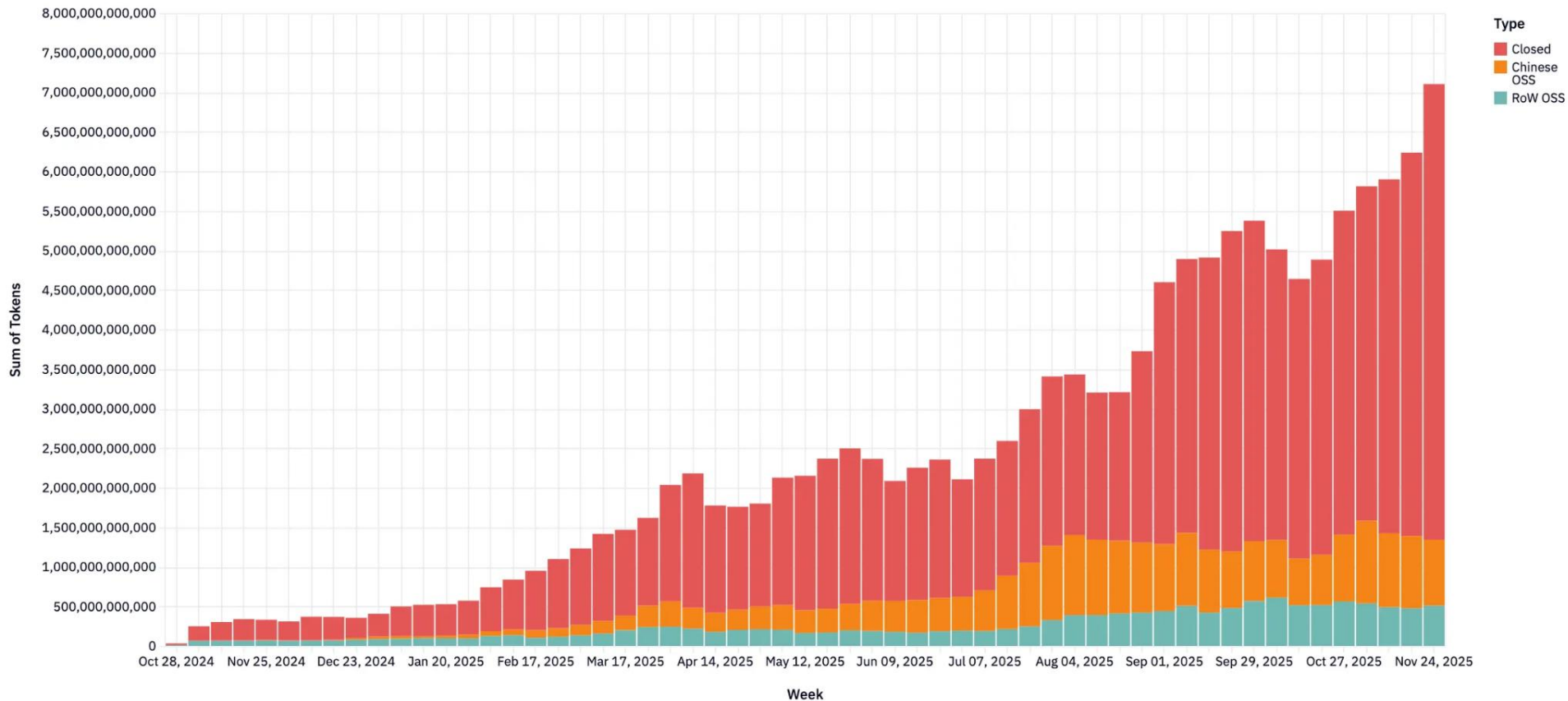
Contest Session

April 26, 2026

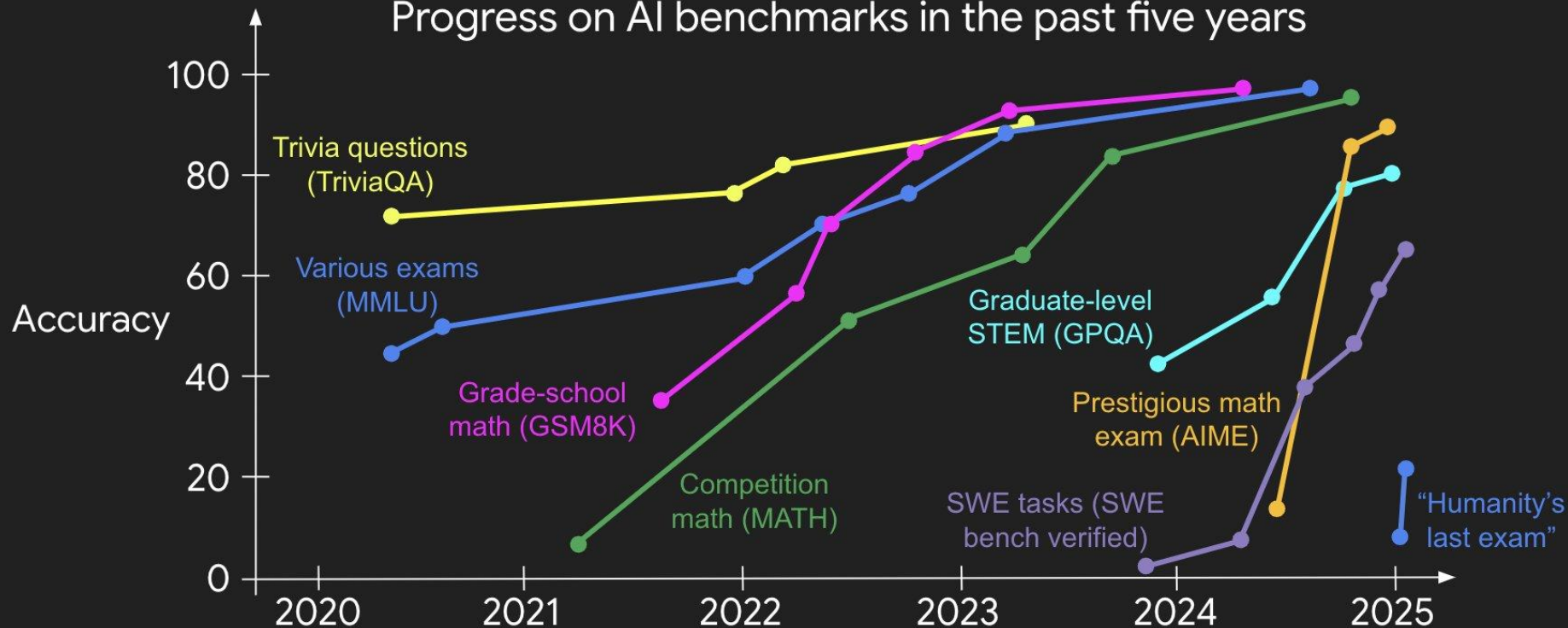
problem set contains 13 problems and spans 20 pages.

ID	English Title	Chinese Title
A	White Night	白夜
B	Yearning for Yonder 2	对远方的向往 2
C	Four Kuhn's Theorem 2	四次伯氏定理 2
D	Call You With Your Name	呼唤你
E	Call You With Your Name 2	呼唤你 2
F	Many Many Heads 2	很多很多头 2
G	Puzzle: The Artisan of Glimoth	谜题：格里莫斯工匠
H	Strange Sorting 2	奇怪的排序 2

Token Usage By Source Type



Progress on AI benchmarks in the past five years





Dillon Uzar
@DillonUzar



Context Arena Update: Added kimi-linear-48b-a3b-instruct [11-08] and kimi-k2 (Thinking) [11-06] to the MRCR leaderboards.

The Linear 48b results are fascinating! It actually outperforms the new Gemini 3.0 Pro Thinking on 4-needle and 8-needle tasks at higher context lengths (512k+). I've added it to 2needle, 4needle, and 8needle.

kimi-k2 (Thinking) lands lower on the leaderboards (Rank #22 for 2-needle AUC @ 128k), with a hard context ceiling around 262k. I did not run it for 2needle and 4needle.

All results at: contextarena.ai

The performance curve for the Linear model is distinct: while it underperforms Gemini 3 significantly at shorter contexts (<=256k) on the difficult 8-needle test, its degradation slope is much flatter. Gemini starts higher and drops fast; Kimi starts lower but holds steady, overtaking Gemini at the higher end.

However, note that kimi-linear-48b has noticeable performance drops past 128k on the easier 2 & 4 needle tests. Additionally, due to lower token efficiency compared to Gemini/GPT, only ~60% of the 1M token tests successfully ran (hitting limits/OOM). So some caution with the results at the 1M level.

kimi-linear-48b results:

kimi-linear-48b results:

2-Needle Performance (@ 128k / @ 1M):

- AUC: 96.5% (vs Gem 3: 99.5%) / 81.7% (vs Gem 3: 85.5%)
- Pointwise: 96.0% (vs Gem 3: 99.0%) / 77.0% (vs Gem 3: 72.2%)

4-Needle Performance (@ 128k / @ 1M):

- AUC: 85.5% (vs 85.8%) / 62.7% (#1, beating Gem 3: 57.3%)
- Pointwise: 83.7% (vs 80.8%) / 51.5% (#1, beating Gem 3: 34.3%)

8-Needle Performance (@ 128k / @ 1M):

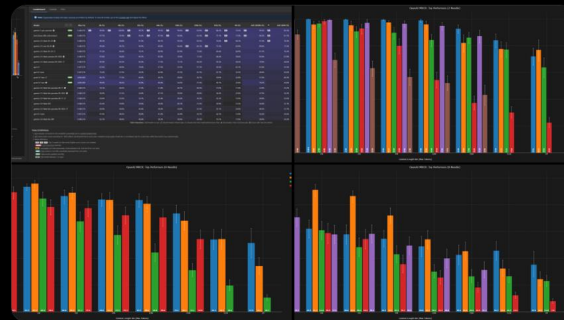
- AUC: 54.9% (vs 73.0%) / 43.8% (#1, beating Gem 3: 39.0%)
- Pointwise: 49.0% (vs 54.2%) / 35.3% (#1, beating Gem 3: 24.5%)

A very different architectural approach yielding impressive stability at scale. Because of its current price point, it is very competitive for long context (MRCR).

Enjoy.

@Kimi_Moonshot
@GoogleDeepMind @googleaidevs
@OpenAI @OpenAIDevs

翻译帖子



上午3:34 · 2025年11月23日 · 27.3万 查看

21

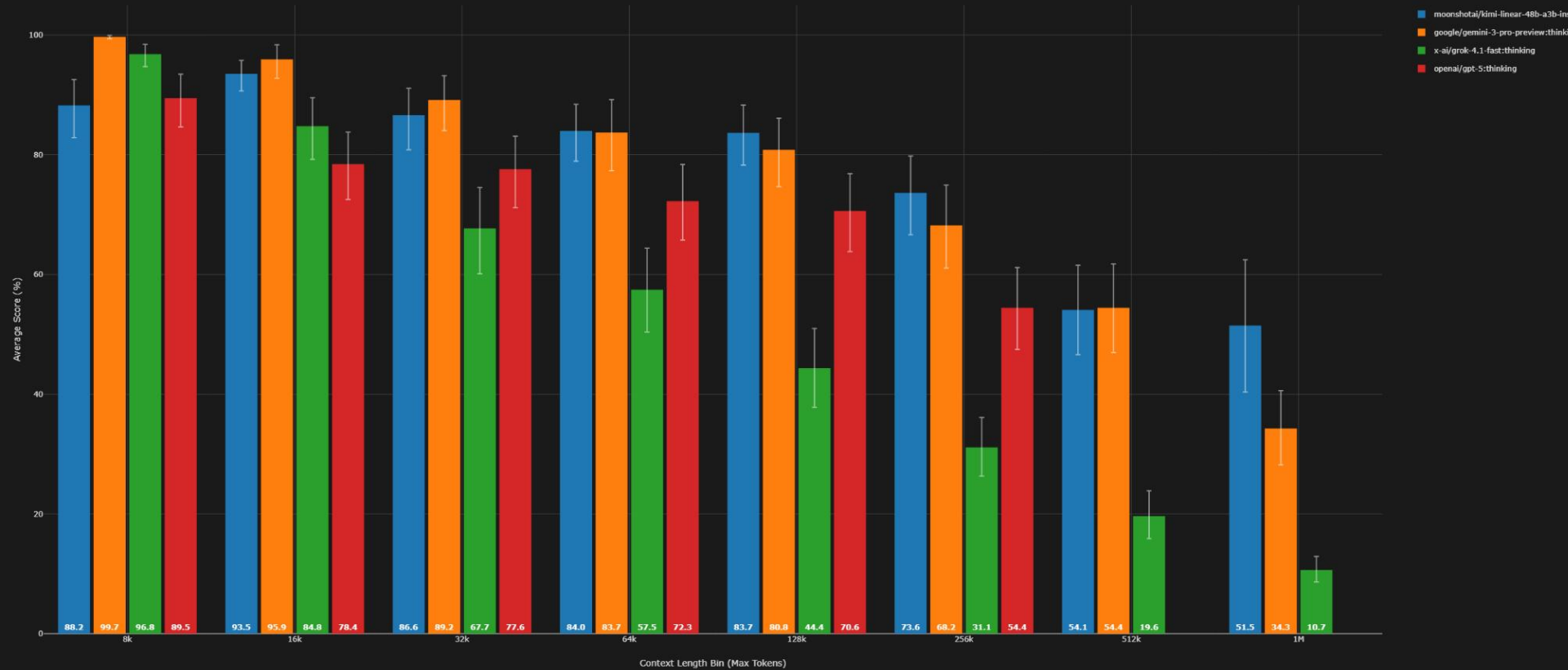
72

462

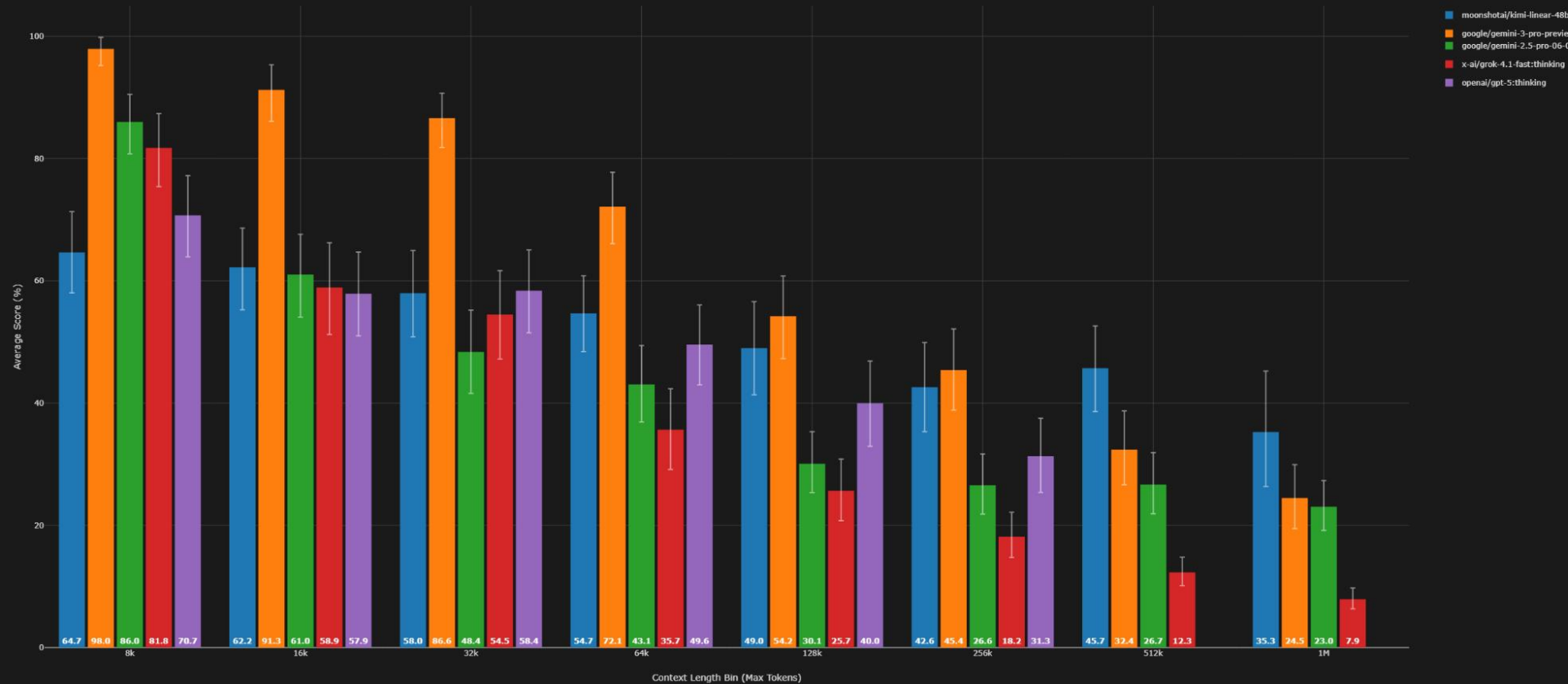
233



OpenAI-MRCR: Top Performers (4-Needle)



OpenAI-MRCR: Top Performers (8-Needle)



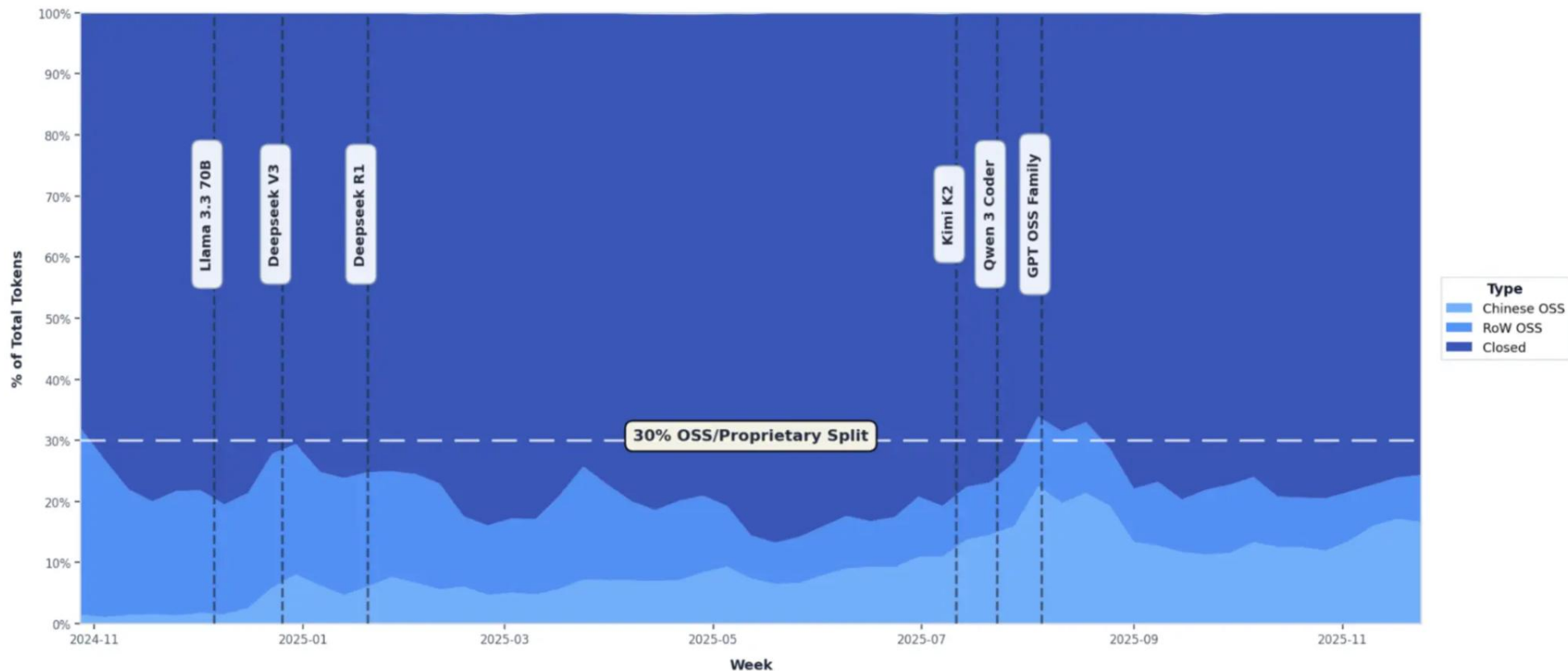


Moonshot /

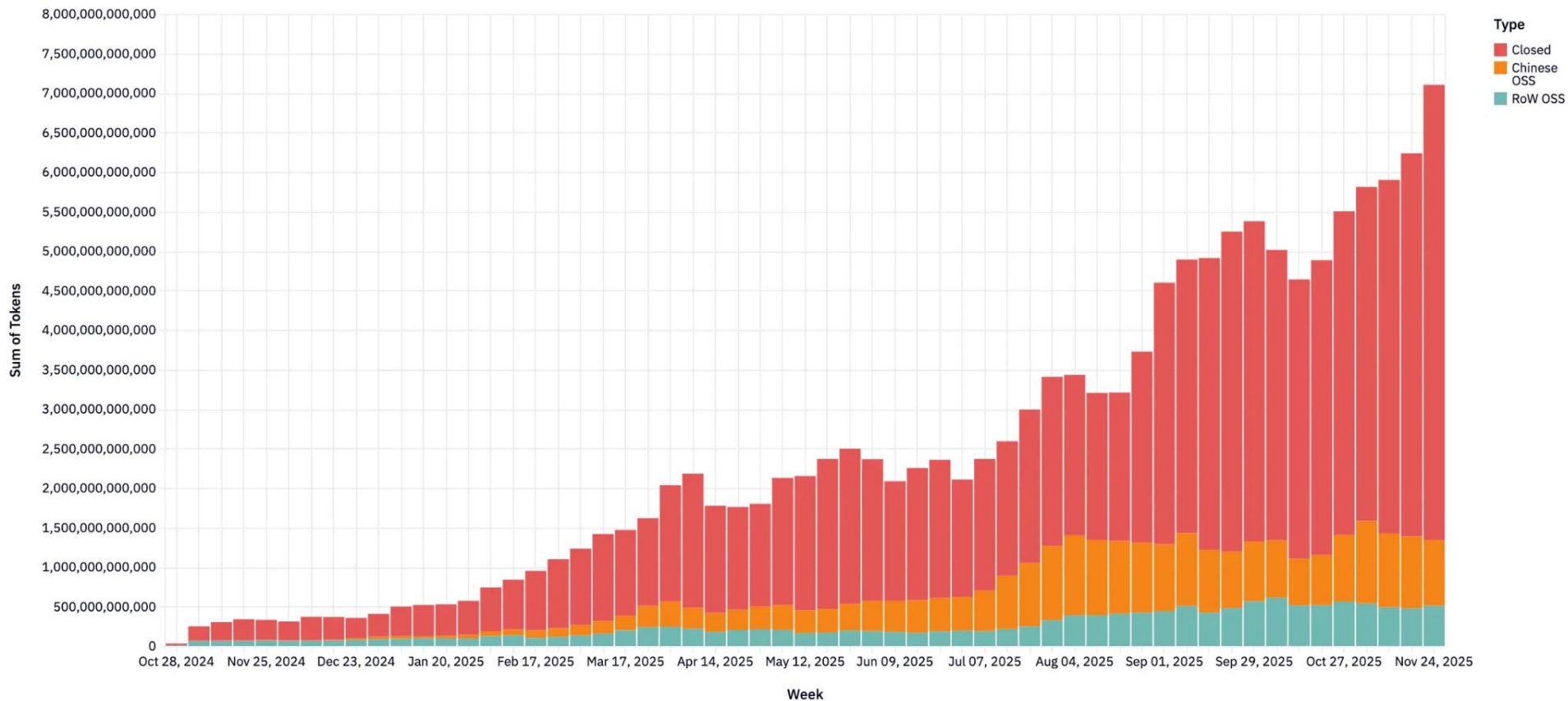
Kimi Linear



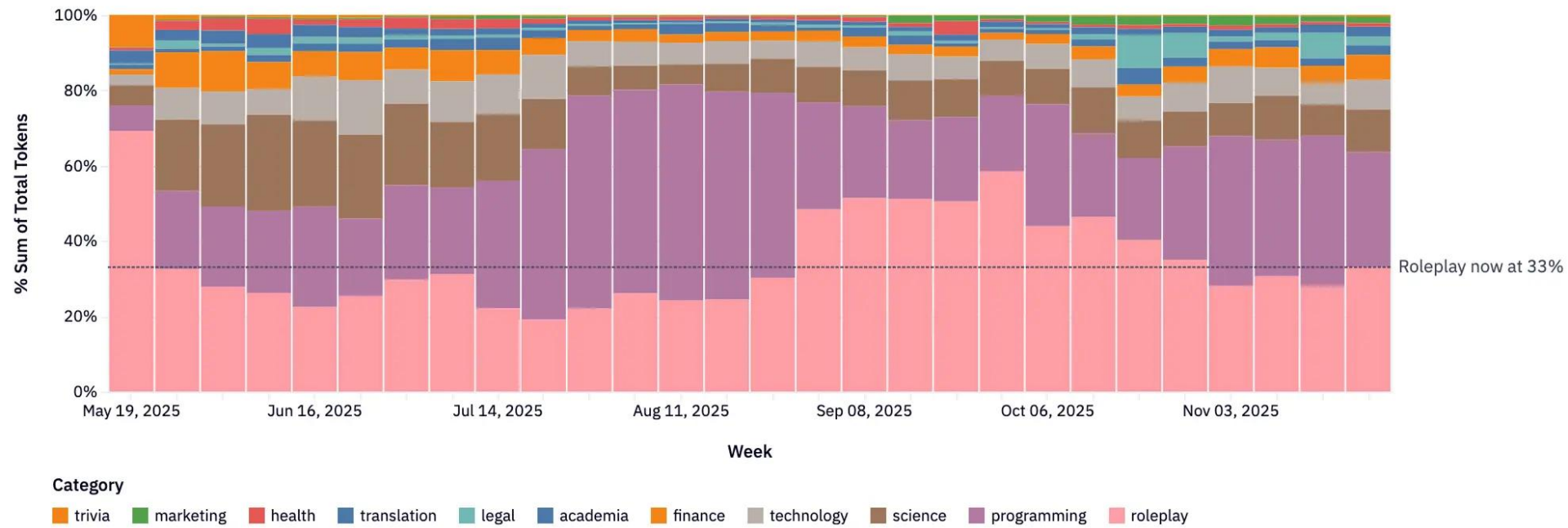
Token Share by Source Type and Origin



Token Usage By Source Type



Category Breakdown of Chinese OSS Models



Kimi For Coding 权益

尝鲜体验

Andante

¥49/月

重度开发

Allegretto

¥199/月

日常使用

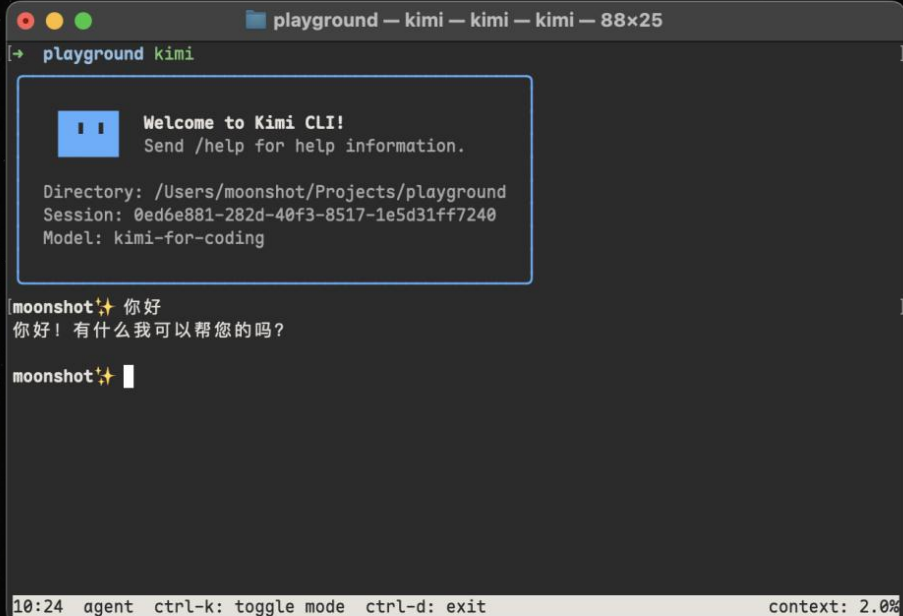
Moderato

¥99/月

与 Claude Code 套餐相比, **1/7** 价格, 至高 **3X** 用量
极速体验 无需等待

仅限个人开发者使用, 禁止共享账号

KIMI





Code Categories Arena



DesignArena by Arcada Labs

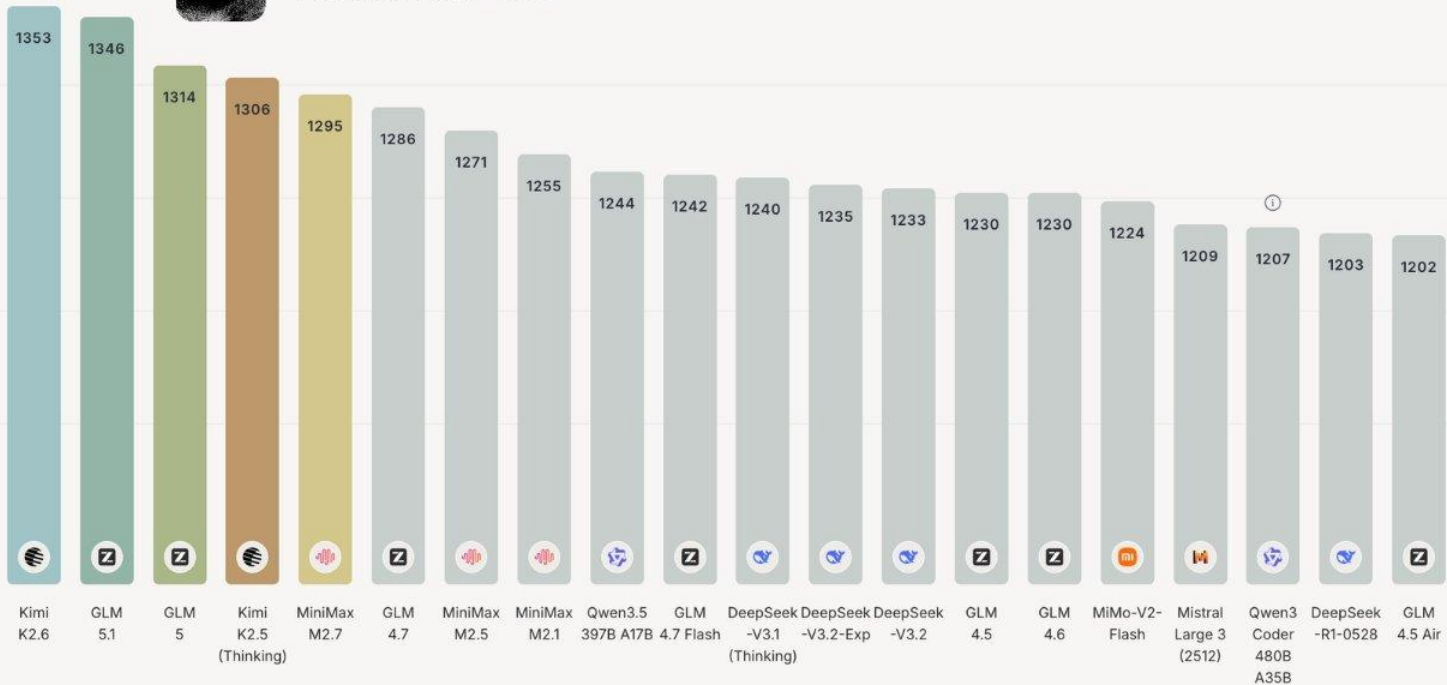
Open Weights ▾

Elo Rating

Win Rate



Kimi K2.6



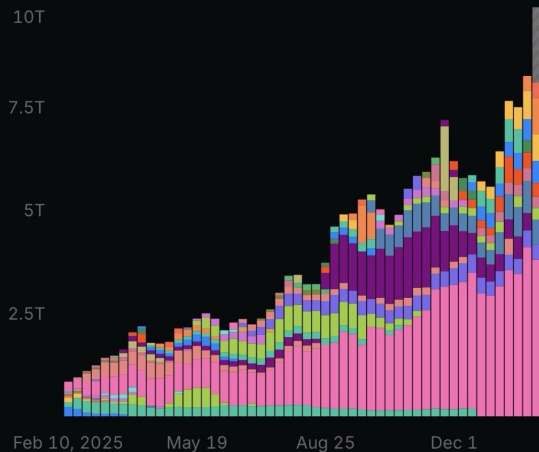


Kimi K2.6 by Moonshot AI

is now on Design Arena!



DesignArena by Arcada Labs



LLM Leaderboard

This Week

1. **Kimi K2.5**
by [moonshotai](#) 856B tokens
↑311%
2. **Gemini 3 Flash Preview**
by [google](#) 796B tokens
↑32%
3. **Claude Sonnet 4.5**
by [anthropic](#) 711B tokens
↓6%
4. **DeepSeek V3.2**
by [deepseek](#) 626B tokens
↑31%

Kimi.ai 已转帖

**Kimi Product**
@KimiProduct**OpenClaw**

X

x

**Kimi**

Kimi x OpenClaw — The Simplest Setup Yet

💬 76

↺↻ 336

❤️ 3,222

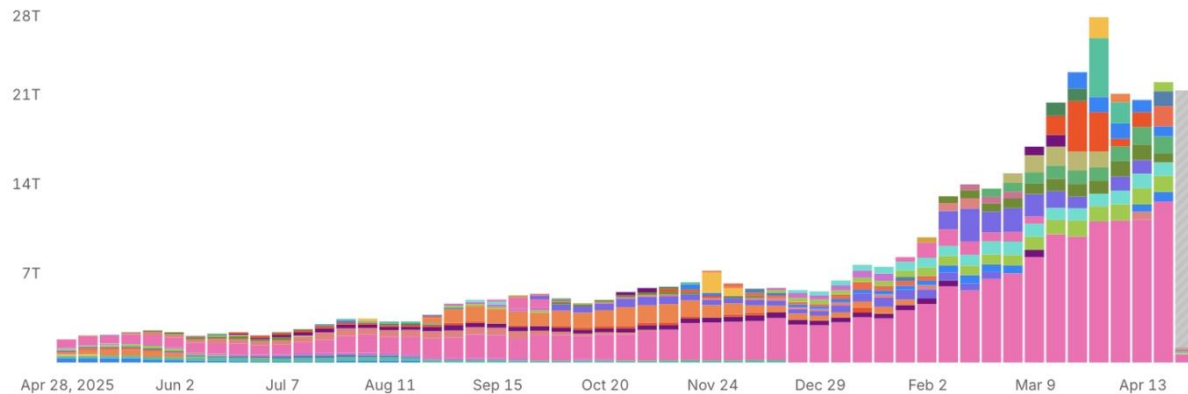
📊 30万



Kimi K2.5 is currently the #1 most-used model for OpenClaw on OpenRouter!

Top Models

Weekly usage of models across OpenRouter



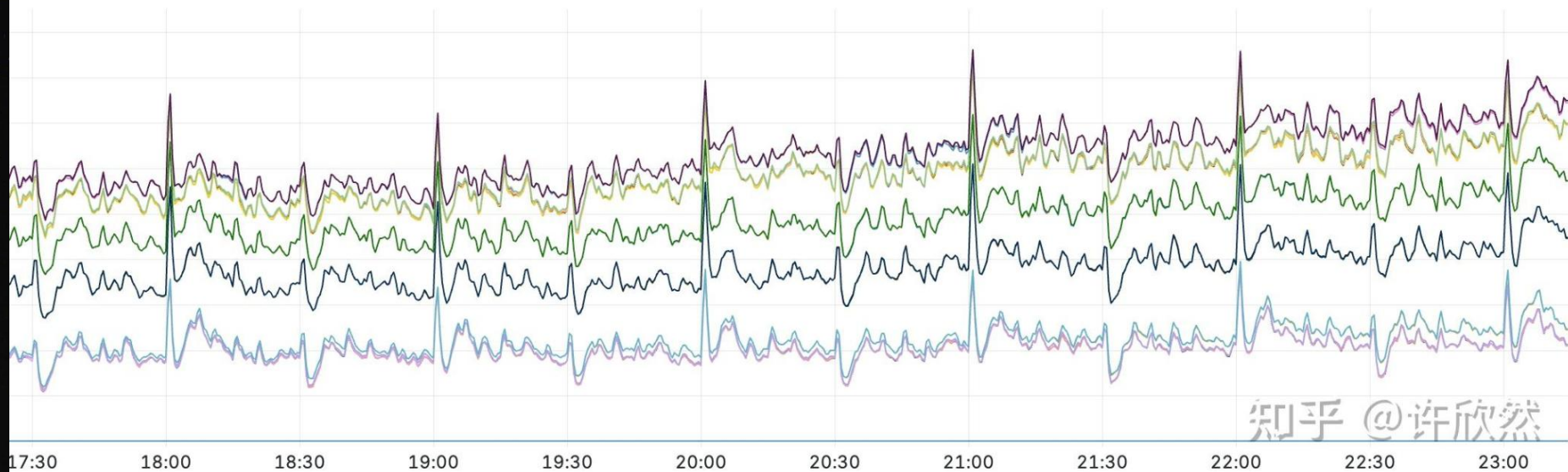
LLM Leaderboard

Compare the most popular models on OpenRouter [?](#)

This Week

1.	 Kimi K2.6 by moonshotai	1.58T tokens new	6.	 MiniMax M2.7 by minimax	790B tokens ↓ 18%
2.	 Claude Sonnet 4.6 by anthropic	1.36T tokens ↓ 2%	7.	 Grok 4.1 Fast by x-ai	747B tokens ↑ 30%
3.	 DeepSeek V3.2 by deepseek	1.28T tokens ↓ 0%	8.	 Step 3.5 Flash by stepfun	721B tokens ↑ 98%
4.	 Claude Opus 4.7 by anthropic	1.15T tokens ↑ 279%	9.	 Claude Opus 4.6 by anthropic	685B tokens ↓ 44%
5.	 Gemini 3 Flash Preview by google	1.04T tokens ↓ 9%	10.	 Gemini 2.5 Flash Lite by google	651B tokens ↑ 9%

RPM





“We believe everyone has been happily raising their lobsters(openclaw), but when setting heartbeating jobs, please avoid scheduling on the hour(e.g. 10:00/10:30) scheduled tasks, the surge every hour of openclaw requests poses a huge challenge to our inference services. It feels like swarms of lobsters attacking the Moon via space elevator every hour(with reinforcements arriving half an hour later).”



杨攀同学

Entrepreneur, Pixel-Perfect

02/11

如果你知道今年 GPU 算力的总产能，
如果你了解今年 Token 消耗的双指数
增长逻辑，那么，你一定会得出一个确
定性的结论：

2026 年 Token 要涨价 💰 💰 💰

长按扫码
查看详情



帖子



DHH

@dhh



更多

So impressed with Kimi K2.5 for Omarchy development. I made four changes: Default enable Hibernation, Add "Remove > Preinstalls" (here's your bare-bones install!), documenting all bin/*, fix gum confirm color problem. Total tokens: 116K. Cost: \$0.63. Via OpenCode / Zen. So good.

[翻译帖子](#)

下午6:11 · 2026年2月7日 · 12.5万 查看



53



50



1,214



338



相关 ∨

[查看引用 >](#)



发布你的回复

回复



DHH

@dhh · 2月7日



In fact, I prefer it to Opus for this simple stuff because it's a lot faster. So the feedback loop is just nicer. It's crazy to see that the open-weight models have gotten this good!



13



1



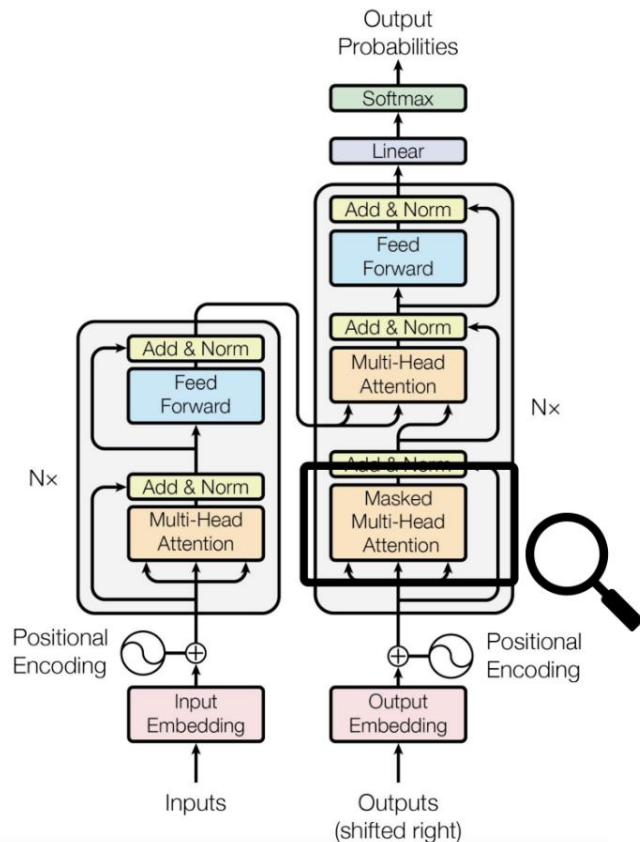
340



3万



Attention in Transformers [Vaswani et al. '17]



Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

Lukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Attention: Training

L : sequence length

d : hidden state dimension

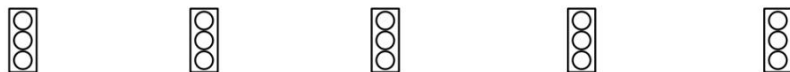
$$\mathbf{O} \in \mathbb{R}^{L \times d}$$



$$\mathbf{O} = \text{SelfAttention}(\mathbf{X})$$



$$\mathbf{X} \in \mathbb{R}^{L \times d}$$



Attention: Training

L : sequence length

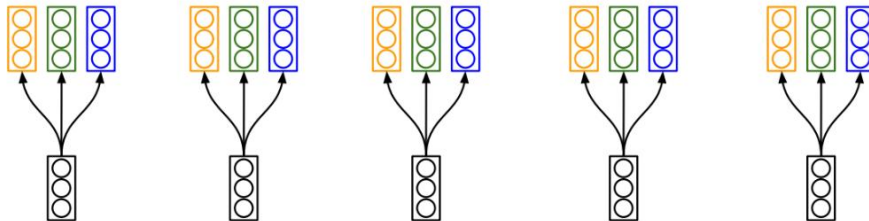
d : hidden state dimension

$O(Ld^2)$

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} = \mathbf{XW}_Q, \mathbf{XW}_K, \mathbf{XW}_V$$

$$\mathbf{X} \in \mathbb{R}^{L \times d}$$

Key K
Value V
Query Q



Attention: Training

L : sequence length

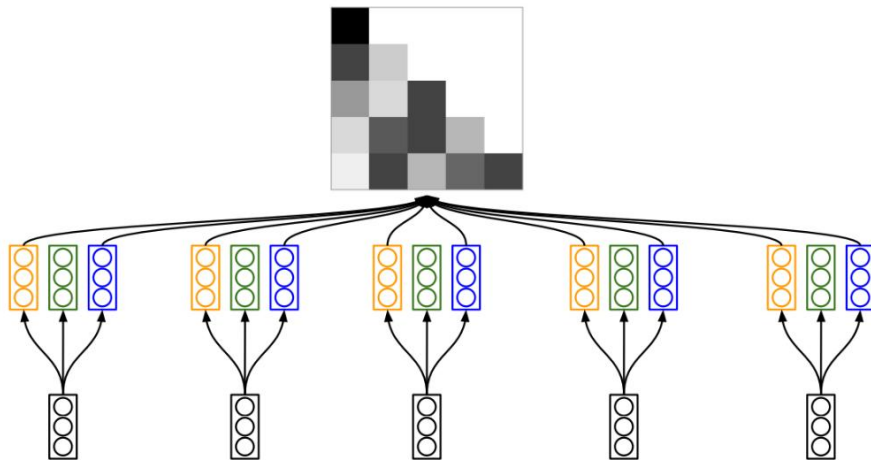
d : hidden state dimension

$$O(L^2 d) \quad \mathbf{A} = \text{softmax}(\mathbf{Q}\mathbf{K}^\top \odot \mathbf{M}) \in \mathbb{R}^{L \times L}$$

$$O(Ld^2) \quad \mathbf{Q}, \mathbf{K}, \mathbf{V} = \mathbf{X}\mathbf{W}_Q, \mathbf{X}\mathbf{W}_K, \mathbf{X}\mathbf{W}_V$$

$$\mathbf{X} \in \mathbb{R}^{L \times d}$$

Key K
Value V
Query Q



Attention: Training

L : sequence length

d : hidden state dimension

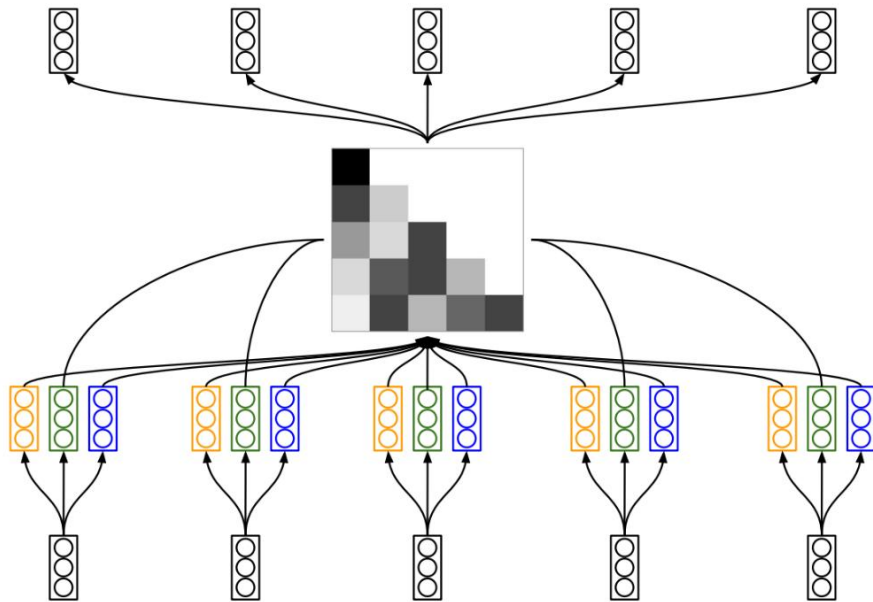
$$O(L^2 d) \quad \mathbf{O} = \mathbf{A}\mathbf{V} \in \mathbb{R}^{L \times d}$$

$$O(L^2 d) \quad \mathbf{A} = \text{softmax}(\mathbf{Q}\mathbf{K}^T \odot \mathbf{M}) \in \mathbb{R}^{L \times L}$$

$$O(Ld^2) \quad \mathbf{Q}, \mathbf{K}, \mathbf{V} = \mathbf{X}\mathbf{W}_Q, \mathbf{X}\mathbf{W}_K, \mathbf{X}\mathbf{W}_V$$

$\mathbf{X} \in \mathbb{R}^{L \times d}$

Key	K
Value	V
Query	Q



Attention

	Training (“Parallel Form”)	Inference (“Recurrent Form”)
	$\mathbf{O} = \text{softmax}((\mathbf{Q}\mathbf{K}^\top) \odot \mathbf{M})\mathbf{V}$	$\mathbf{o}_t = \frac{\sum_{i=1}^t \exp(\mathbf{q}_t \mathbf{k}_i^\top) \mathbf{v}_i}{\sum_{i=1}^t \exp(\mathbf{q}_t \mathbf{k}_i^\top)}$
Compute	$O(L^2)$ ☹️	$O(L^2)$ ☹️
Memory	$O(L)$ 😊	$O(L)$ ☹️
Steps	$O(1)$ 😊	$O(L)$

Attention enables scalable training of accurate sequence models, but requires:

- Quadratic compute for training.
- Linear memory for inference.

From Softmax to Linear Attention [Katharopoulos et al. '20]

Softmax
Attention

$$\mathbf{O} = \cancel{\text{softmax}}((\mathbf{Q}\mathbf{K}^\top) \odot \mathbf{M})\mathbf{V}$$

$$\{-\infty, 0\}^{L \times L}$$

(Simple) Linear
Attention

$$\mathbf{O} = ((\mathbf{Q}\mathbf{K}^\top) \odot \mathbf{M})\mathbf{V}$$

$$\{0, 1\}^{L \times L}$$

From Softmax to Linear Attention [Katharopoulos et al. '20]

Training (“Parallel Form”)

Inference (“Recurrent Form”)

Softmax
Attention

$$\mathbf{O} = \text{softmax}((\mathbf{Q}\mathbf{K}^\top) \odot \mathbf{M})\mathbf{V}$$

$$\mathbf{o}_t = \sum_{j=1}^t \frac{\exp(\mathbf{q}_t^\top \mathbf{k}_j)}{\sum_{l=1}^t \exp(\mathbf{q}_t^\top \mathbf{k}_l)} \mathbf{v}_j$$

(Simple) Linear
Attention

$$\mathbf{O} = ((\mathbf{Q}\mathbf{K}^\top) \odot \mathbf{M})\mathbf{V}$$

$$\mathbf{o}_t = \sum_{j=1}^t (\mathbf{q}_t^\top \mathbf{k}_j) \mathbf{v}_j$$

Linear Attention: Naive Parallel Form

Training (“Parallel Form”)

$$\mathbf{O} = ((\mathbf{Q}\mathbf{K}^\top) \odot \mathbf{M})\mathbf{V}$$

Inference (“Recurrent Form”)

$$\mathbf{o}_t = \sum_{j=1}^t (\mathbf{q}_t^\top \mathbf{k}_j) \mathbf{v}_j$$

Compute

$$O(L^2) \quad \text{☹️}$$

$$O(L) \quad \text{😊}$$

Memory

$$O(L) \quad \text{😊}$$

$$O(1) \quad \text{😊}$$

Steps

$$O(1) \quad \text{😊}$$

$$O(L)$$

Linear attention has constant-memory inference, but still requires:

- Quadratic compute for training.
- (Can theoretically use recurrent form + parallel scan for $O(L)$ compute and $O(\log L)$ work, but this is not at all practical.)

THE BOTTLENECK

Transformer's Quadratic Dilemma

Standard Attention Computation

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T/\sqrt{d})V$$

Computing QK^T generates an $N \times N$ matrix, causing memory and computational complexity to grow quadratically with sequence length

Complexity Analysis

Memory Complexity

$$O(N^2)$$

Computational Complexity

$$O(N^2d)$$

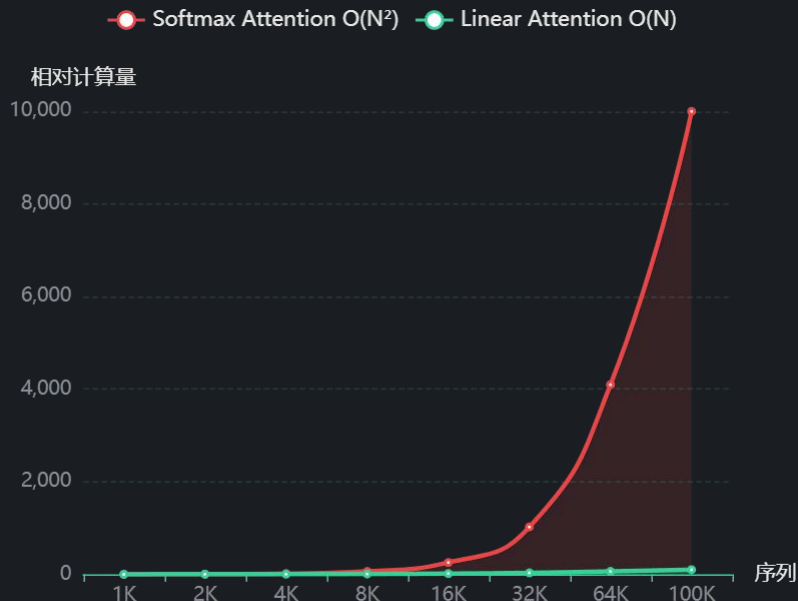
When Sequence Length $N=100K$

10 Billion Operations

Practical Application Limits

When processing long documents, high-resolution images, or videos, quadratic complexity causes GPU memory shortage and slow computation, severely limiting Transformer's applicability

Computational Complexity Comparison



BREAKTHROUGH

The Birth of Linear Attention: From Theory to Practice

Core Innovation

Katharopoulos et al. (2020) proposed replacing Softmax with **kernel functions** to replace Softmax, utilizing **Matrix Multiplication Associativity** to change computation order

$$\text{sim}(\mathbf{Q}, \mathbf{K}) = \varphi(\mathbf{Q}) \cdot \varphi(\mathbf{K})^T$$

Computation Flow Comparison

Traditional Attention

$$(\varphi(\mathbf{Q})\varphi(\mathbf{K})^T) \cdot \mathbf{V} \rightarrow O(N^2d)$$

Linear Attention

$$\varphi(\mathbf{Q}) \cdot (\varphi(\mathbf{K})^T \cdot \mathbf{V}) \rightarrow O(Nd^2)$$

Key Advantages

- Constant Memory:** only maintain a fixed-size state during inference
- Linear Complexity:** computational cost grows linearly with sequence length
- Parallel Training:** maintains Transformer's parallelization advantages

Linear Attention Computation Flow

Q

- Kernel Function Mapping**
 $\varphi(\mathbf{Q}) = \text{kernel}(\mathbf{Q})$

K,V

- Precompute KV State**
 $\mathbf{S} = \varphi(\mathbf{K})^T \cdot \mathbf{V}$

Out

- Linear Attention Output**
 $\mathbf{O} = \varphi(\mathbf{Q}) \cdot \mathbf{S} / \text{normalization}$

Core Insight: By changing computation order, complexity is reduced from $O(N^2d)$ to $O(Nd^2)$, achieving orders-of-magnitude efficiency gains in long-sequence scenarios

PERFORMANCE GAP

The Performance Gap Mystery: Why Linear Attention Underperformed

Core Problem

Despite linear attention's significant efficiency advantages, early approaches on vision and language tasks had **accuracy significantly lower than** Softmax attention, limiting their practical applications

Injectivity Loss

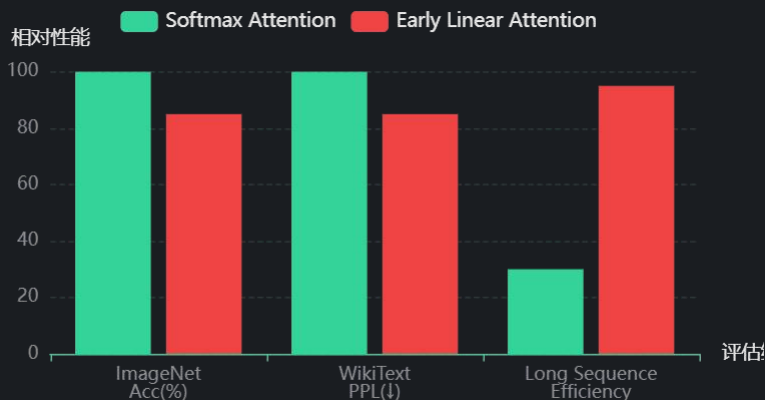
Tsinghua University Research (2024) discovered: Softmax attention is **an injective function**, while linear attention is **not**

Consequence: different queries may produce identical attention distributions, causing **semantic confusion**, the model cannot distinguish certain semantics

Insufficient Local Modeling Ability

Softmax attention's success depends not only on long-range modeling, **effective local modeling** is equally crucial. Linear attention typically produces flatter distributions, lacking the sharp focusing ability to capture local details

Performance Comparison Illustration



Other Limitations

- Insufficient Feature Diversity:** linear attention may overly smooth feature representations
- Unstable Gradients:** unbounded gradients affect training stability
- Attention Dilution:** weight distribution too dispersed

- **The Evaluation Trap:
Goodhart's Law in Action**
- **Infrastructure: Where
Theory Meets Metal**



CHAPTER 02

Linear Attention's Core Principles

Reveal the mathematical principles and algorithmic mechanisms behind
linear attention

The Magic of Kernel Functions and Matrix Multiplication

Kernel Function Mapping

Through carefully designed **kernel function** ϕ , map queries Q and keys K to a new feature space, making similarity computation decomposable

Original Form

$$\text{sim}(Q, K) = \exp(QK^T)$$

Kernel Form

$$\text{sim}(Q, K) = \phi(Q)\phi(K)^T$$

Matrix Multiplication Associativity

Key insight: utilizing $(AB)C = A(BC)$ to change computation order, avoiding building $N \times N$ matrices

Traditional: compute QK^T first

$$O(N \cdot d \cdot N) = O(N^2d)$$

Linear: compute K^TV first

$$O(d \cdot N \cdot d) = O(Nd^2)$$

Common Kernel Functions

$$\phi(x) = \text{elu}(x) + 1$$

- Exponential Linear Unit

$$\phi(x) = \text{ReLU}(x)$$

- Rectified Linear Unit

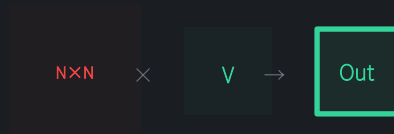
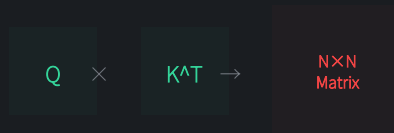
$$\phi(x) = \text{softmax}(x)$$

- Applied separately to Q, K

Computation Order Transformation Illustration

Traditional Attention

$O(N^2d)$



$O(Nd^2)$

COMPLEXITY COMPARISON

Complexity Comparison: The Leap from $O(N^2)$ to $O(N)$

Softmax Attention

Time

$O(N^2d)$

Space

$O(N^2)$

Bottleneck: QK^T matrix requires N^2 memory

Linear Attention

Time

$O(Nd^2)$

Space

$O(d^2)$

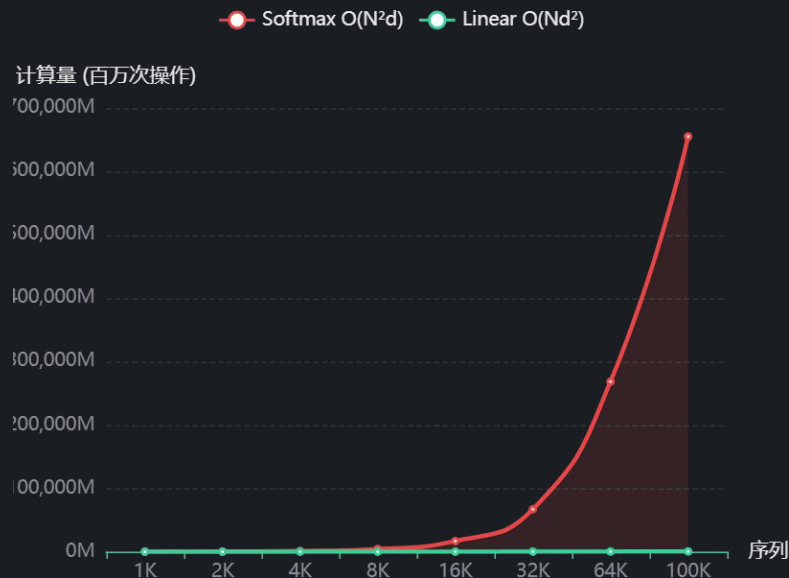
Advantage: only maintain $d \times d$ state matrix

Key Assumption

Linear attention's advantages manifest when $N \gg d$. Typical values: when $N=4096$, $d=64$,

$N/d=64\times$ gap

Actual Computation Comparison ($d=64$)



Linear Attention's Evolution Path

Track important models and milestones from the birth to maturity of linear attention

Development Timeline

1

Linear Transformer (2020)

First proposed linear attention, using feature maps to replace Softmax

2

Performer (2021)

Adopted Orthogonal Random Features (PORF) to approximate Softmax

3

Efficient Attention (2021)

Applied Softmax separately to Query and Key

4

GLA (2023)

Introduced data-dependent gating mechanisms, performance rivaling Softmax

5

Focused Linear Attention (2023)

Solved focusing ability and feature diversity issues

MILESTONE 01

Linear Transformer: Pioneering Work

Paper Information

Title: Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention

Authors: Katharopoulos et al.

Conference: ICML 2020

Citations: 1500+

Core Contributions

- First systematically introduced **kernel methods** into attention mechanisms
- Proved that Transformers can be viewed as **Recurrent Neural Networks (RNN)**
- Achieved **$O(N)$** complexity autoregressive generation

Technical Details

$$\phi(x) = \text{elu}(x) + 1$$

Using ELU+1 as kernel function, ensuring positivity

Linear Transformer Architecture

Input

Linear Attention Layer

$$\begin{aligned} S_i &= S_{i-1} + \phi(k_i)^T v_i \\ z_i &= \phi(q_i) \cdot S_i \end{aligned}$$

State

Accumulated State Matrix

$S \in \mathbb{R}^{(d \times d)}$ recursively updated over time, no need to store historical KV

Output

Linear Attention Output

Each token inference only requires $O(d^2)$ computation and memory

Historical Significance: Opened a new direction in linear attention research, proving Transformers can achieve linear complexity through mathematical transformations, laying the theoretical foundation for subsequent research

Performer: Breakthrough with FAVOR+ Algorithm

Paper Information

Title: Rethinking Attention with Performers

Authors: Choromanski et al. (Google Research)

Conference: ICML 2021

Citations: 2000+

FAVOR+ Algorithm

Fast Attention Via Orthogonal Random features

Utilizing **Orthogonal Random Features** to approximate Softmax kernel functions

Unbiased Estimation: theoretically unbiased approximation of Softmax

Orthogonal Features: reduce variance, improve approximation quality

Performance

In **long-sequence tasks** approaches Softmax performance

In **protein sequence modeling** achieves SOTA on tasks

Supports **ultra-long sequences** (100K+ tokens)

FAVOR+ Algorithm Principles

1. Softmax Kernel Approximation

$$\exp(QK^T/\sqrt{d}) \approx \Phi(Q)\Phi(K)^T$$

$$\Phi(x) = \exp(-||x||^2/2) \cdot [\exp(\omega_1^T x), \dots, \exp(\omega_m^T x)]$$

2. Orthogonal Random Features

Using **orthogonalMatrix** to generate random features, reducing variance compared to independent sampling, improving approximation accuracy

3. Linear Complexity Implementation

$O(N^2d)$

$O(Nmd)$

$m \ll N$ Achieves significant acceleration when

Core Innovation: Through random feature approximation, achieves linear complexity while maintaining Softmax expressiveness, providing a practical solution for long-sequence modeling

Mamba: A New Chapter in State Space Models

Paper Information

Title: Mamba: Linear-Time Sequence Modeling with Selective State Spaces

Authors: Gu & Dao (2023)

Citations: 3000+

Impact: triggered an SSM research boom

Core Innovation

Selective Mechanism: parameters B, C, Δ become functions of input

Hardware-Aware Algorithm: parallel scan enables efficient training

Linear Complexity: $O(N)$ training, $O(1)$ inference

Performance Breakthrough

In language modeling matches Transformer++

Long-sequence modeling significantly outperforms traditional attention

Inference speed 5-10x improvement

Mamba Selective SSM

1. Continuous-time SSM

$$\mathbf{h}'(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{h}(t)$$

2. Discretization & Selectivity

Discretization: using Zero-Order Hold (ZOH)

$$\tilde{\mathbf{A}} = \exp(\Delta\mathbf{A}), \tilde{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$$

Selectivity: $B, C, \Delta = \text{Linear}(x)$

making parameters input-dependent, enabling content-aware processing

3. Parallel Scan Algorithm

Through associative scan, completes sequence computation in $O(\log N)$ parallel steps, balancing training parallelism and inference efficiency

Paradigm Shift: Mamba proved state space models can match or surpass Transformers, opening a new era of non-attention architectures, inspiring extensive follow-up research

Focused Linear Attention: Enhanced Focusing Ability

Paper Information

Title: FLatten Transformer: Vision Transformer using Focused Linear Attention

Authors: Han et al. (Tsinghua University, 2023)

Conference: ICCV 2023

Problem Analysis

Insufficient Focusing Ability: linear attention distribution too flat

Missing Feature Diversity: leading to decreased expressiveness

Insight: Softmax's sharp distribution is key to its success

Solutions

Focused Function (fp): enhances attention sharpness

Depthwise Separable Convolution (DWC): complements local modeling

Focused Linear Attention Module

1. Focused Function

$$fp(x) = \text{sign}(x) \cdot |x|^p, 0 < p < 1$$

Compresses large values and amplifies small values via power functions, making attention distribution sharper, simulating Softmax's focusing effect

2. Depthwise Separable Convolution (DWC)

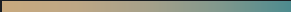
Introduces DWC branch in parallel, enhancing local feature extraction, compensating for linear attention's local modeling deficiency

3. Rank Recovery Module

Restores attention matrix rank through simple linear transformations, enhancing feature diversity

Performance Breakthrough: On DeiT-Tiny, Top-1 accuracy improved from 72.2% to 74.1% (+1.9%), proving linear attention can surpass Softmax attention through simple improvements

KDA: Kimi Delta Attention Architecture Analysis



The Perfect Fusion of Fine-Grained Gating and Hardware Efficiency



Channel-wise Fine-Grained Gating

KDA introduces **channel-wise gating mechanisms**, each feature dimension maintains independent forgetting rates, achieving more precise finite-state RNN memory regulation compared to head-wise gating.

GDN/Mamba2: Head-wise Gating (Coarse-grained)

KDA: Channel-wise Gating (Fine-grained)

Advantage: More precise memory management, stronger expressiveness

Chunkwise-Parallel Hardware Optimization

KDA customizes dedicated chunkwise-parallel algorithms, optimizing memory access patterns and parallel strategies for efficient training on modern GPUs, surpassing FlashAttention-2 even for short sequences.

- Optimized memory access patterns
- Efficient block-level parallel computation
- Seamless integration with vLLM

DPLR Matrix Parameterization

KDA adopts **Diagonal-Plus-Low-Rank (DPLR)** matrix special variant to parameterize transition dynamics, significantly reducing computation while maintaining classical delta rule consistency.

DPLR = Diagonal + Low-Rank

Diagonal part captures main dynamics, low-rank part models complex interactions

Kimi Linear Hybrid Architecture

KDA and full attention layers are interleaved at **3:1 ratio** evenly, maintaining linear attention's efficiency while preserving global information flow through full attention layers.

75%

Memory Reduction

6×

Throughput Improvement

Decoding Performance at 1M Context Length

AttnRes: Attention Residual Mechanism Revolution



A Paradigm Breakthrough of "Attentionizing" Depth Dimension



PreNorm Dilution Problem

Traditional residual connections use **fixed unit-weight accumulation**, causing:

Hidden State Growth: magnitude grows with depth $O(L)$

Layer Contribution Dilution: early information gets submerged

Training Instability: deep layer outputs too large

Core Insights

Treat **depth dimension** as "time", replace fixed accumulation with softmax attention, achieving "on-demand review" instead of "full stacking".

$$h_l = \sum \alpha_{\{i \rightarrow l\}} \cdot v_i$$

α computed via softmax, each layer equipped with learnable pseudo-query vectors

Block AttnRes Chunking Strategy

Divide L layers into **N chunks (about 8)**, standard residual accumulation within chunks, attention aggregation between chunks.

Memory	$O(Ld) \rightarrow O(Nd)$
Complexity:	$< 2\%$
Inference Latency	$1.25\times$
Increase: Compute	

Equivalent:

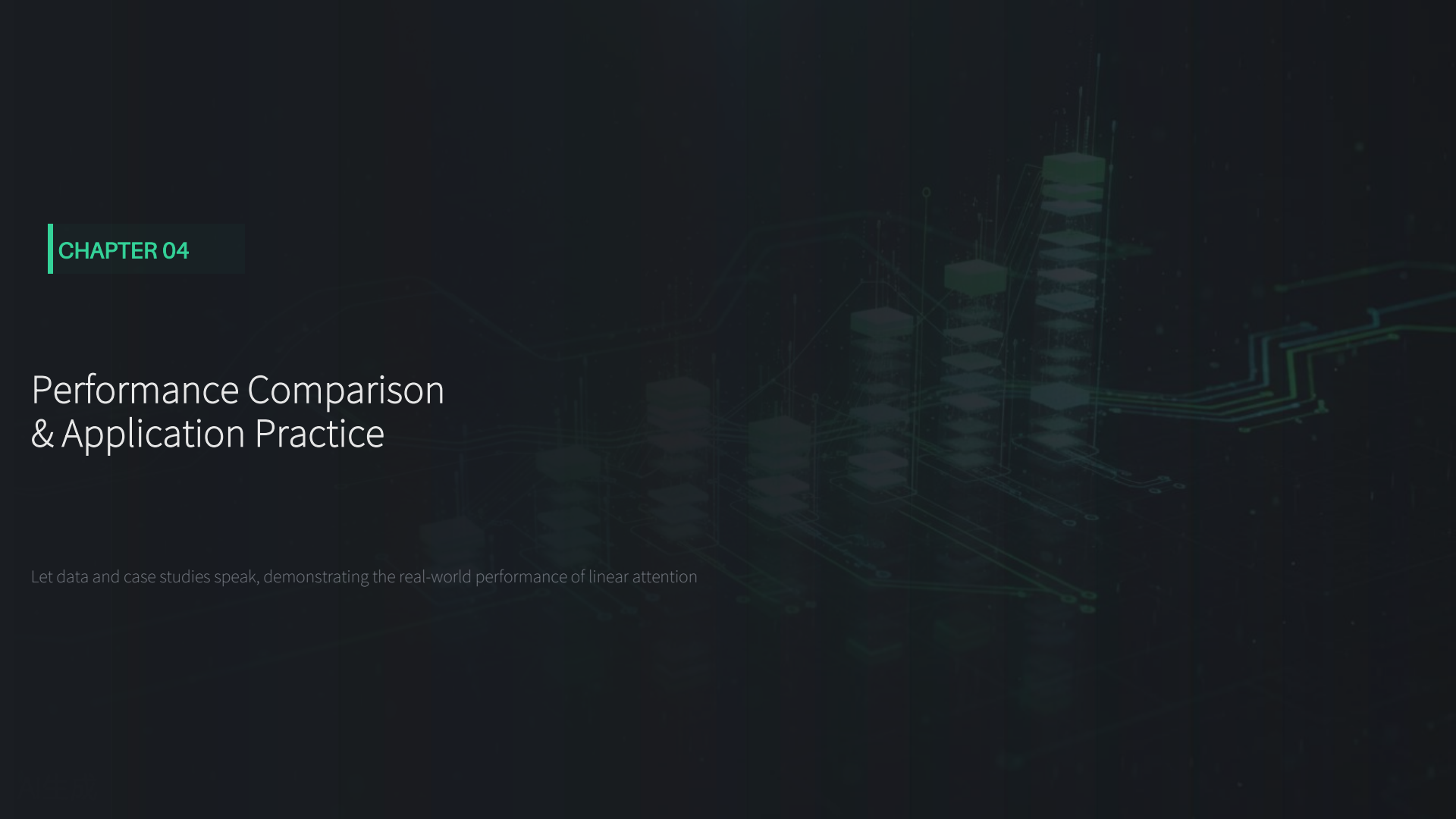
Experimental Validation Results

GPQA-Diamond	
Multi-step Reasoning Benchmark	+7.5min

HumanEval	
Code Generation Benchmark	+3.1min

Kimi Linear 48B parameters, 1.4T tokens pre-training

Paradigm Significance: AttnRes completes the "Attentionization" of the depth dimension, equivalent to doing another Transformer between layers, hailed by researchers as "Deep Learning 2.0" architectural revolution.



CHAPTER 04

Performance Comparison & Application Practice

Let data and case studies speak, demonstrating the real-world performance of linear attention

LONG SEQUENCE MODELING

Long-Sequence Modeling: Overwhelming Efficiency Advantage

RAT: 10x Acceleration

100K Sequence: 7x training acceleration

Inference Stage: 10x speed improvement

KV Cache: significantly reduced, lowering OOM risk

GPU Kernel Optimization

3.3x Acceleration: compared to SOTA linear attention implementation

3.6x Memory Reduction: optimized data movement

Surpasses FlashAttention-2: faster when $N > 3000$

Efficiency Comparison

Sequence Length

16K

32K

64K

100K

Speedup Ratio

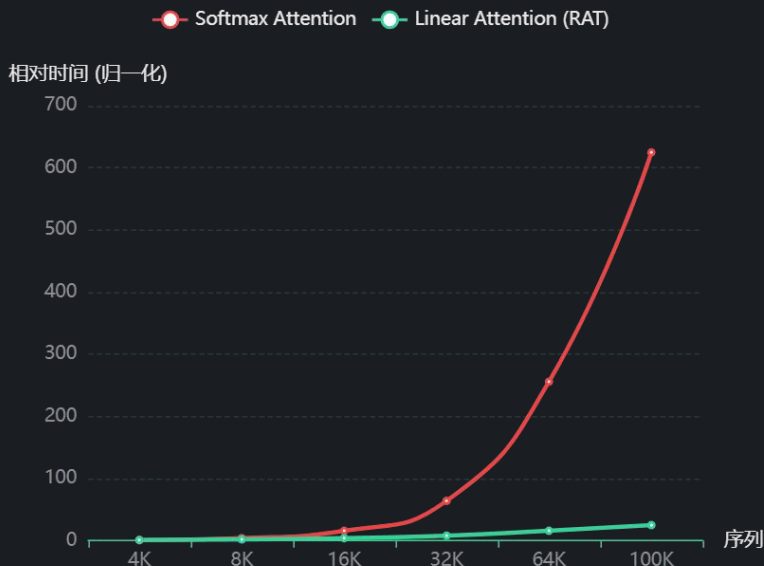
2×

3×

4×

7×

Long-Sequence Efficiency Comparison



Multimodal Application: Successful Practice in CLIP

Research Background

In **Multimodal Transformers**, visual and text tokens total thousands, quadratic complexity becomes a severe bottleneck. Linear attention provides a natural solution for efficient multimodal learning

Experimental Setup

Dataset: LAION-400M

Model: ViT-S/B/L-16

Framework: OpenCLIP

Comparison: Linear Attention vs Standard Attention

Key Findings

Comparable Performance: Linear attention CLIP

achieves standard attention performance

Scaling Laws: Follows similar accuracy-model size scaling laws

Efficiency Advantage: significantly reduces training and inference costs

CLIP Architecture & Linear Attention Application

Vision Encoder

ViT

Image Tokens

$N=197$

Linear Attention

$O(N)$

Text Encoder

Transformer

Text Tokens

$N=77$

Linear Attention

$O(N)$

Contrastive Learning Objective

$$L = -\log(\exp(\text{sim}(I, T)) / \sum_j \exp(\text{sim}(I, T_j)))$$

Practical Significance: Proves the feasibility of linear attention in multimodal representation learning, providing an efficient path for large-scale multimodal model training, especially advantageous for high-resolution images and long texts



Kimi Linear实验结果与技术趋势

🏆 Comprehensive Performance Evaluation

After 1.4T tokens pre-training, Kimi Linear (48B total/3B active parameters) **matches or surpasses** strong full-attention baselines on short-context, long-context, and RL-style post-training tasks.

+7.5

GPQA-Diamond
Multi-step Reasoning Benchmark

+3.1

HumanEval
Code Generation Benchmark

🔑 效率提升

Long-Sequence Memory Reduction

KV-cache优化

75%

Decoding Throughput Improvement

1M上下文长度

6×

Inference Latency Increase

Block AttnRes

<2%

🔮 Future Technology Trends

1

Finer-Grained Gating Mechanisms

Explore element-wise or even dynamic gating

2

Deep Integration with SSM

Combine advantages of Mamba and other state space models

3

Ultra-Long Context Optimization

Efficient processing for million-scale token scenarios

4

Cross-modal scenarios: vision, audio, etc.

5

Scaling Law Impact



Open Source Ecosystem: KDA kernel and vLLM integration are open-sourced, pre-trained models and instruction-tuned checkpoints available for research and studies.

The core of Kimi Linear is Kimi Delta Attention (KDA) — an improved version of Gated DeltaNet (GDN) that introduces more efficient gating mechanisms to optimize finite-state RNN memory usage. The authors note that while GDN, similar to Mamba2, uses coarse head-wise forget gates, KDA introduces a channel-wise variant where each feature dimension maintains independent forgetting rates, similar to Gated Linear Attention (GLA).

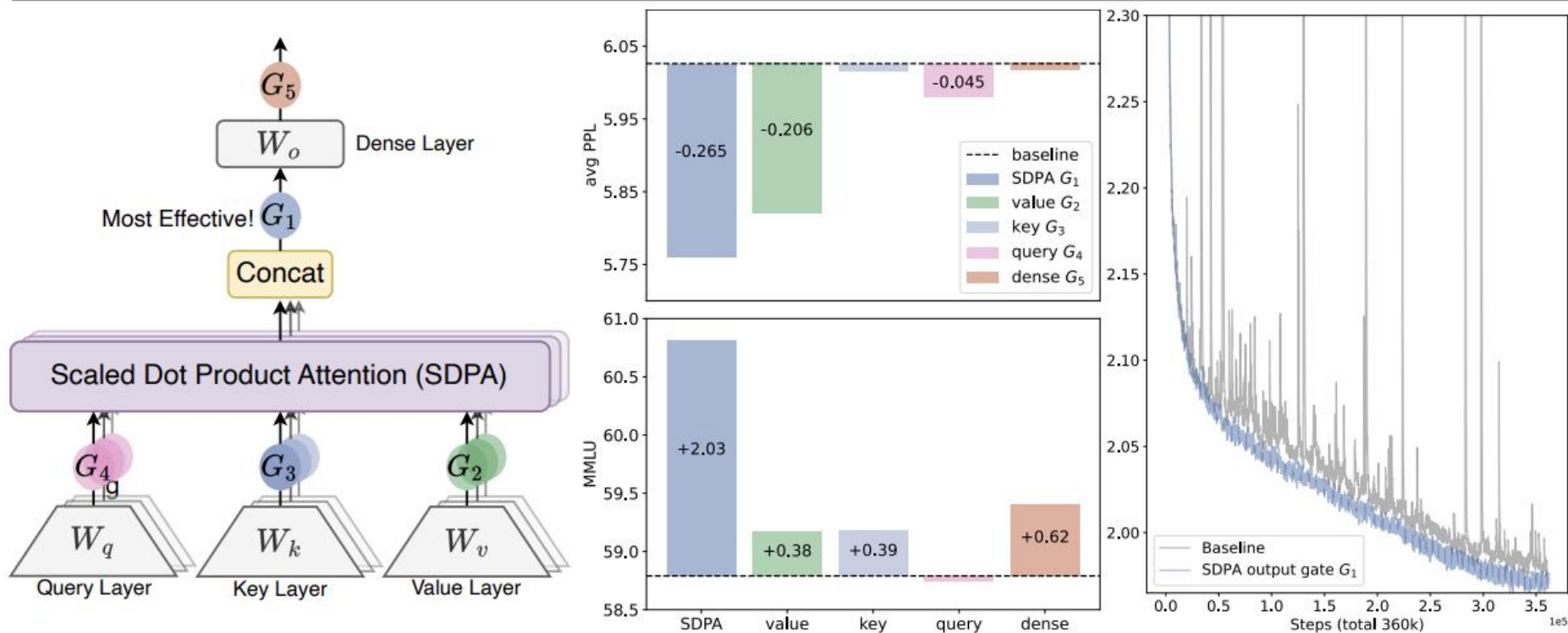


Figure 1: Left: Investigated positions for applying gating operations.; **Middle:** Performance comparison (Test PPL and MMLU) of 15B MoE models with gating applied at various positions. Gating after SDPA (G_1) yields the best overall results. Gating after the Value layer (G_2) also demonstrates notable improvements, particularly in PPL. **Right:** Training loss comparison (smoothed, 0.9 coeff.) over 3.5T tokens between baseline and SDPA-gated 1.7B dense models under identical hyperparameters. Gating results in lower final loss and substantially enhanced training stability, mitigating loss spikes. This stability allows for potentially higher learning rates and facilitates better scaling.

This fine-grained design enables more precise regulation of finite-state RNN memory, unlocking the potential of RNN-style models in hybrid architectures. The key is that KDA parameterizes its transition dynamics through a specialized variant of Diagonal-Plus-Low-Rank (DPLR) matrices, enabling a custom chunked parallel algorithm that significantly reduces computation compared to generic DPLR formulations while remaining consistent with the classical delta rule. Kimi Linear interleaves KDA with periodic full attention layers at a fixed 3:1 ratio.

The authors pre-trained a Kimi Linear model based on a layered hybrid architecture of KDA and Multi-Head Latent Attention (MLA). The model has 3B active parameters and 48B total parameters.

CORE INSIGHTS

Core Insights

Linear attention is transitioning from **theoretical exploration** to **engineering practice**

KDA and AttnRes represent key milestones in this evolution



Fine-Grained Gating

Channel-wise Memory
Management



Hardware Optimization

Chunkwise-parallel
Algorithms



Deep Attentionization

AttnRes Paradigm
Revolution

We are witnessing **yet another paradigm shift** in Transformer architecture

CHAPTER 05

Future Outlook & Challenges

Look ahead to the future development directions and challenges facing linear attention

FUTURE DIRECTIONS

Frontier Directions: Sparse State Expansion & Hybrid Architectures

Hardware Co-optimization: Unleashing Linear Attention's Potential

Unsolved Mysteries: Retrieval Tasks & In-Context Learning

etcs...

Computer Science > Distributed, Parallel, and Cluster Computing

[Submitted on 16 Apr 2026 (v1), last revised 22 Apr 2026 (this version, v2)]

Prefill-as-a-Service: KVCache of Next-Generation Models Could Go Cross-Datcenter

Ruoyu Qin, Weiran He, Yaoyu Wang, Zheming Li, Xinran Xu, Yongwei Wu, Weimin Zheng, Mingxing Zhang

Prefill-decode (PD) disaggregation has become the standard architecture for large-scale LLM serving, but in practice its deployment boundary is still determined by KVCache transfer. In conventional dense-attention models, prefill generates huge KVCache traffics that keep prefill and decode tightly coupled within a single high-bandwidth network domain, limiting heterogeneous deployment and resource elasticity. Recent hybrid-attention architectures substantially reduce KVCache size, making cross-cluster KVCache transport increasingly plausible. However, smaller KVCache alone does not make heterogeneous cross-datcenter PD serving practical: real workloads remain bursty, request lengths are highly skewed, prefix caches are unevenly distributed, and inter-cluster bandwidth fluctuates. A naive design that fully externalizes prefill can therefore still suffer from congestion, unstable queueing, and poor utilization.

We present Prefill-as-a-Service (PrfaaS), a cross-datcenter serving architecture that selectively offloads long-context prefill to standalone, compute-dense prefill clusters and transfers the resulting KVCache over commodity Ethernet to local PD clusters for decode. Rather than treating reduced KVCache as sufficient, PrfaaS combines model-side KV efficiency with system-side selective offloading, bandwidth-aware scheduling, and cache-aware request placement. This design removes the requirement that heterogeneous accelerators share the same low-latency RDMA fabric, enabling independent scaling of prefill and decode capacity across loosely coupled clusters. In a case study using an internal 1T-parameter hybrid model, a PrfaaS-augmented heterogeneous deployment achieves 54% higher serving throughput and 64% lower P90 TTFT than a homogeneous PD baseline, with approximately 15% throughput gain at equal cost, while consuming only modest cross-datcenter bandwidth.

Comments: 16 pages, 5 figures, 6 tables

Subjects: **Distributed, Parallel, and Cluster Computing (cs.DC)**

Cite as: [arXiv:2604.15039](https://arxiv.org/abs/2604.15039) [cs.DC]

(or [arXiv:2604.15039v2](https://arxiv.org/abs/2604.15039v2) [cs.DC] for this version)

<https://doi.org/10.48550/arXiv.2604.15039> 

Submission history

THE FUTURE IS LINEAR

Linear Attention's Future is Here

From theoretical breakthroughs to practical applications, linear attention is
reshaping the deep learning landscape

Although challenges remain, its **efficiency**, **scalability** and **expressiveness** continuous progress indicates this technology
will play an increasingly important role in future AI systems

$O(N)$

Linear Complexity
Breaking Computational Bottlenecks

10×

Inference Acceleration
Unlocking Deployment Potential

100K+

Long-Sequence Support
Expanding Application Scenarios

A person is silhouetted against a bright blue background, holding a smartphone up. The phone's screen displays the text "Scale up!". The person's face is not visible, and the background has a soft, out-of-focus quality.

Scale up!



References

- [A Survey of Vibe Coding with Large Language Models](#)
- [Google Engineering Practices Documentation](#)
- [Context-Engineering-for-AI-Agents-Lessons-from-Building-Manus](#)
- [Linear Transformers for Efficient Sequence Modeling](#)
- [Why Did MiniMax M2 End Up as a Full Attention Model?](#)
- [Everything about AI infra](#)
- [The Second Half](#)
- [Attention Residuals 回忆录](#)
- [how anthropic use claude code](#)

Thanks!

